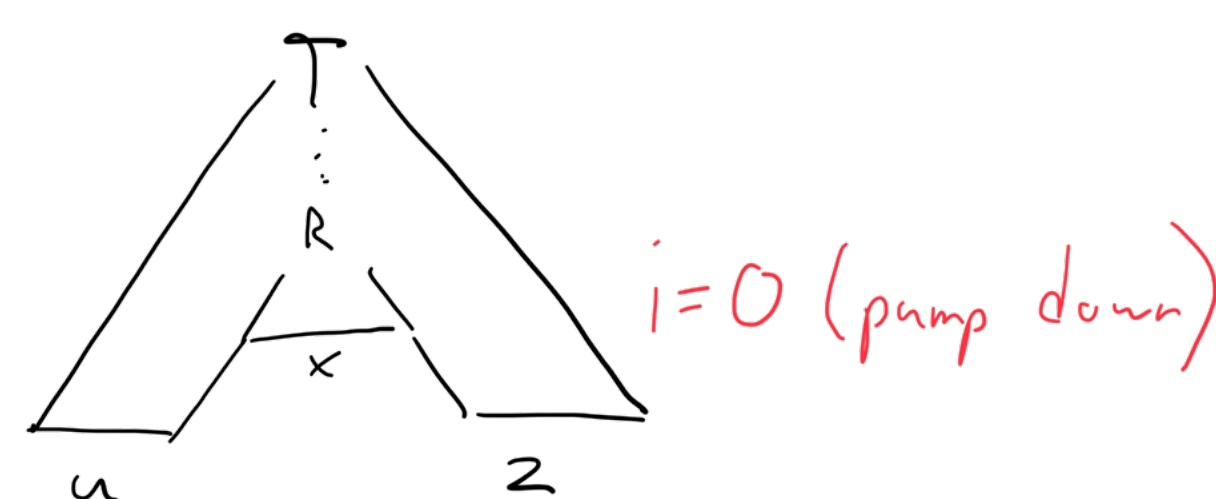
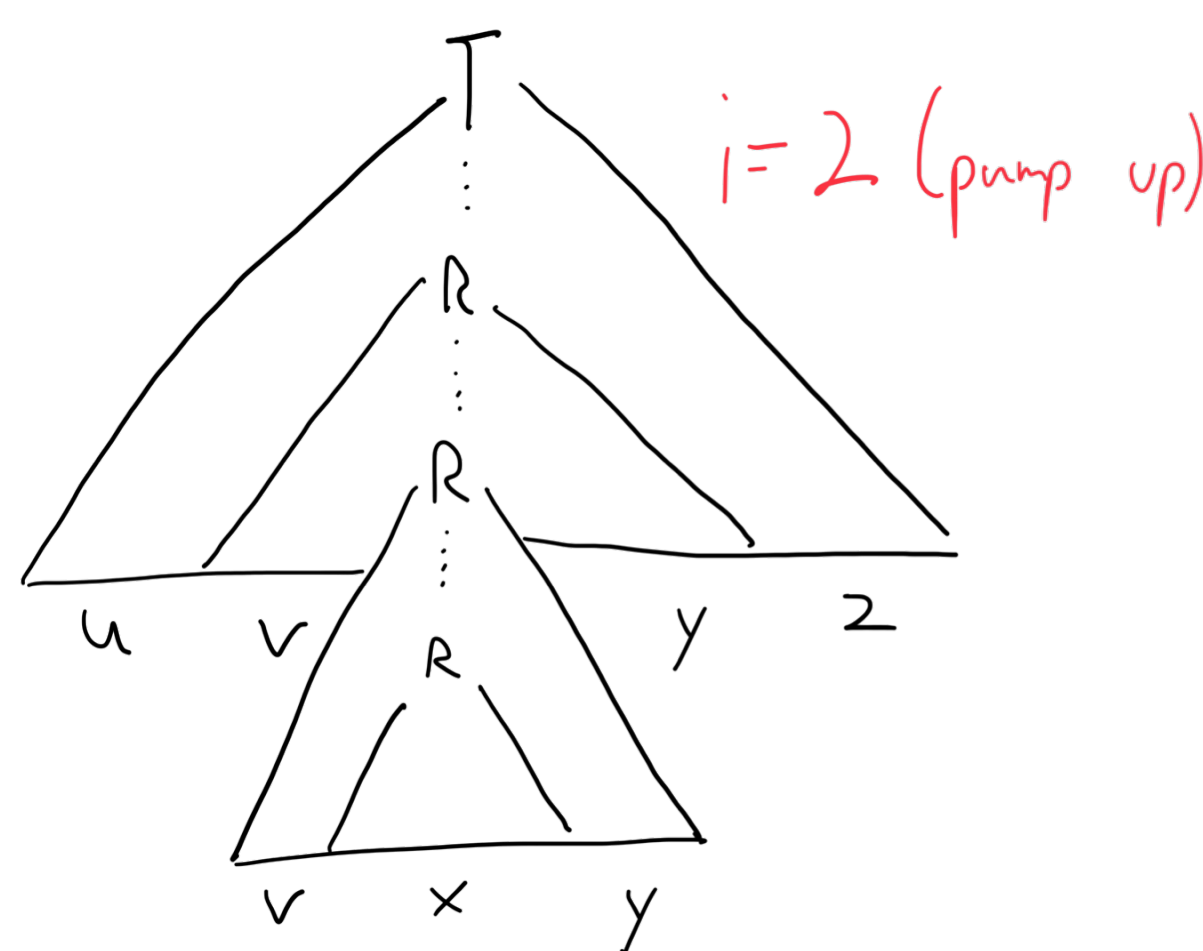
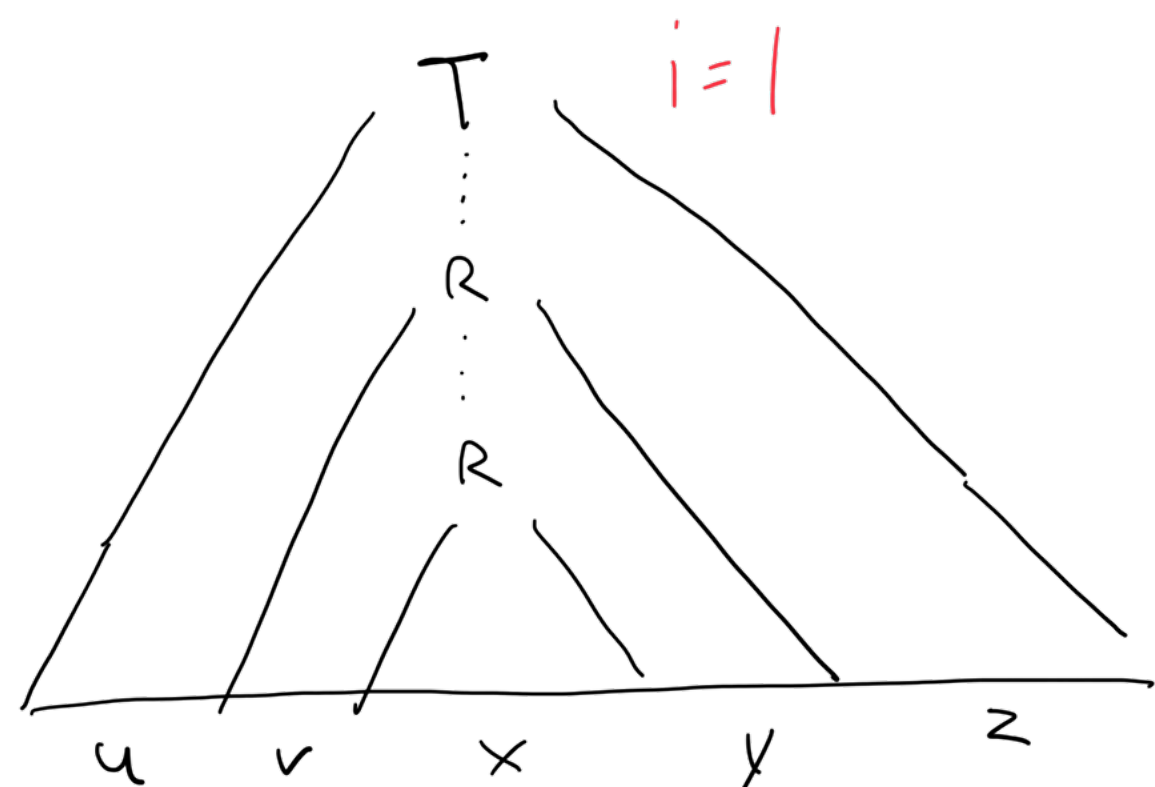


Pumping Lemma for CFL's

Theorem 2.34. If A is a context-free language then there exists a pumping length p where, if s is any string $s \in A$ of length at least p , then s may be divided into five pieces $uvxyz$ satisfying:

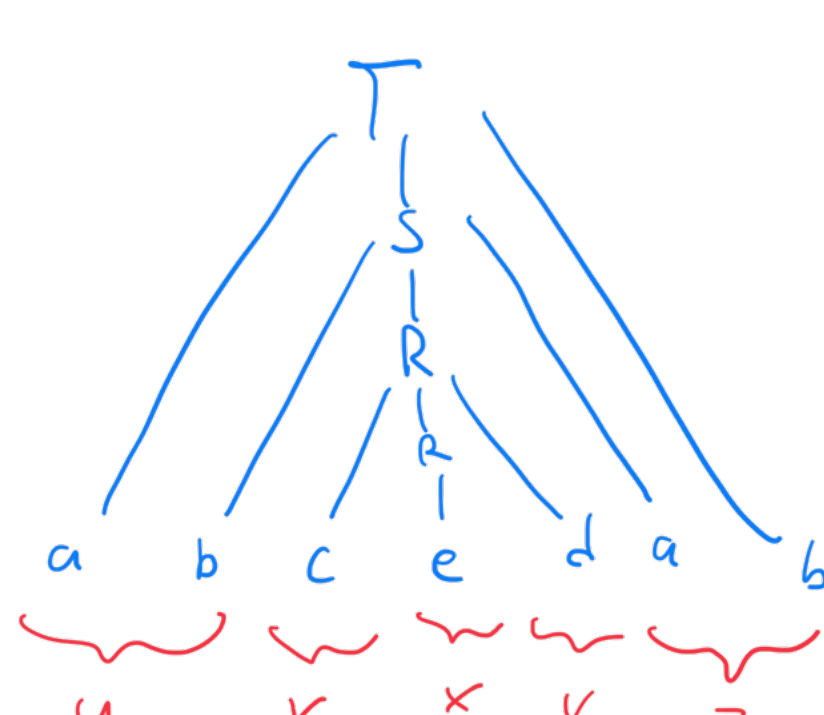
- for each $i \geq 0$, $uv^i x y^i z \in A$.
- $|v y| > 0$ ← either v or y is nonempty
- $|v x y| \leq p$ ← vxy have length at most p

"beginning of v can't be too far from the end of y "



On a sufficiently long $s \in A$, the variable R appears twice on some parse tree. The rules of the grammar are finite but the string is too long (pigeonhole principle). Thus, one R can be replaced with the other R .

$T \rightarrow a S b$
 $S \rightarrow b R a$
 $R \rightarrow c R d \mid e$



$$uxz = abcab \in A$$

$$uv^3xy^3z = abcccddab \in A$$

Ex 1 Show B is not context-free.

$$B = \{a^n b^n c^n \mid n \geq 0\}$$

Proof) Suppose for the sake of contradiction that B is context-free. It has a pumping length p . Let $s = a^p b^p c^p$.

Case 1. v & y contain one type of symbol each.

Thus, $uv^2xy^2z \notin B$ because it does not contain an equal number of a 's, b 's & c 's. a contradiction.

Case 2: Either v or y contains more than one type of symbol without loss of generality. w.l.o.g., suppose that v contains more than one type.

If $ab \in v$, then uv^2xy^2z is not in order (i.e. $\dots abab \dots$)

If $bc \in v$, then uv^2xy^2z is not in order (i.e. $\dots bc bc \dots$)

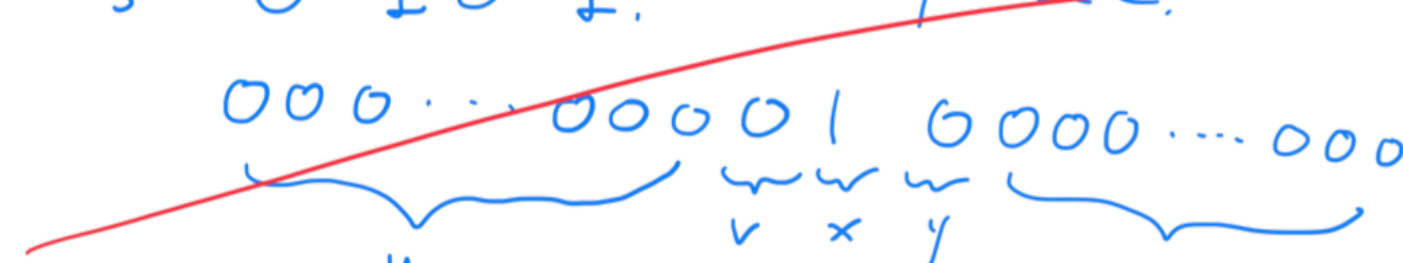
We have a contradiction. $\square$

Ex 2 Let $C = \{ww \mid w \in \{0,1\}^*\}$

Prove it is not a CFL.

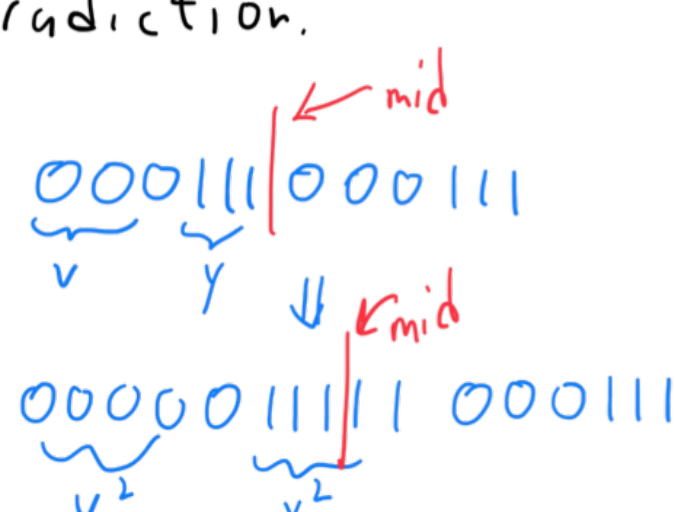
Proof) Suppose C is context-free. It has a pumping length p .

Let $s = 0^p 1 0^p 1$. Clearly $s \in C$.



Let $s = 0^p 1^p 0^p 1^p$.

Case 1: vxy is in the first or second halves of s . wlog, vxy is in the first half. Then, uv^2xy^2z moves a 1 into the second half of s , a contradiction.



Case 2: vxy contains the midpoint of s .



If we pump down, then $uv^0xy^0z = uxz \notin C$ because it gives us $0^p 1^0 0^p 1^p$ where i & j cannot both be p . But, this string is not in the form ww , a contradiction. $\square$